

SPRINGFIELD LINCOLN, IL, USA

WMO: 724390

Lat: 39.845N

Lon: 89.684W

Elev: 594

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	6.2	-7.9	3.7	2.6	-3.1	4.8	7.1	27.5	35.1	25.4	33.4	8.1	280	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	92.8	76.8	90.6	75.9	88.4	74.4	79.8	89.6	78.2	87.5	76.7	85.3	8.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	143.4	86.6	75.3	135.8	84.5	73.9	129.3	82.5	43.7	89.5	42.0	87.6	40.4	85.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.2	19.0	-7.6	96.5	6.6	2.7	-12.4	98.4	-16.3	100.0	-20.0	101.5	-24.8	103.5		
			-8.2	82.1	6.5	1.2	-12.9	83.0	-16.7	83.8	-20.4	84.5	-25.1	85.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	27.3	32.0	42.7	54.3	64.7	73.2	76.1	74.6	67.8	55.9	43.4	32.6	(6)
(7)		DBStd	19.48	12.41	11.88	11.35	9.48	8.58	6.46	5.66	5.39	7.91	9.66	10.25	11.00	(7)
(8)		HDD50	2414	706	515	285	60	3	0	0	0	1	49	248	547	(8)
(9)		HDD65	5284	1168	925	693	340	117	12	1	3	59	313	648	1005	(9)
(10)		CDD50	3815	4	9	60	189	460	697	808	762	536	232	51	7	(10)
(11)		CDD65	1208	0	0	3	18	108	259	344	299	145	31	1	0	(11)
(12)		CDH74	11788	0	0	27	197	971	2489	3609	2769	1443	276	7	0	(12)
(13)		CDH80	4362	0	0	1	38	290	940	1469	1028	531	65	0	0	(13)
(14)	Wind	WSAvg	9.0	10.7	10.7	10.9	11.3	9.3	7.5	6.2	6.0	6.6	8.5	10.2	10.1	(14)
(15)	Precipitation	PrecAvg	35.4	1.6	1.7	3.0	3.8	3.7	3.8	3.7	3.3	3.1	2.8	2.7	2.5	(15)
(16)		PrecMax	52.7	4.5	4.9	7.9	9.9	8.9	9.2	10.7	8.4	8.6	6.4	6.9	8.9	(16)
(17)		PrecMin	25.3	0.0	0.5	1.0	0.7	0.5	0.6	0.9	0.5	0.0	0.2	0.2	0.6	(17)
(18)		PrecStd	6.7	1.1	0.9	1.5	2.2	2.0	2.3	2.2	1.9	2.1	1.7	1.6	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	67.0	77.6	84.5	90.0	93.8	97.9	94.9	93.4	86.2	74.1	64.4	(19)
(20)			MCWB	56.1	57.2	62.6	65.2	72.5	75.8	78.0	76.3	74.0	69.1	60.3	59.3	(20)
(21)		2%	DB	55.6	60.6	71.4	79.8	86.6	91.2	93.0	91.7	89.9	81.1	68.1	58.3	(21)
(22)			MCWB	51.9	53.3	59.4	62.8	69.9	75.2	78.9	76.6	72.5	66.1	58.0	53.1	(22)
(23)		5%	DB	49.7	54.9	66.0	75.1	83.3	88.7	90.4	88.8	86.1	76.5	63.6	53.0	(23)
(24)			MCWB	45.2	48.7	56.6	61.0	68.0	74.3	77.5	75.3	70.5	63.7	56.2	48.8	(24)
(25)		10%	DB	43.5	49.0	60.7	70.7	79.8	85.7	87.7	85.8	82.5	71.8	59.4	48.2	(25)
(26)			MCWB	39.8	43.6	53.1	58.8	66.3	72.7	75.9	73.9	68.5	62.0	53.0	43.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	59.8	64.9	68.7	75.3	79.7	82.1	81.2	77.5	72.8	63.9	60.2	(27)
(28)			MCDB	61.2	64.2	74.4	80.1	86.5	89.3	92.0	90.9	89.5	81.1	69.9	63.8	(28)
(29)		2%	WB	52.6	55.0	61.4	65.7	72.8	77.8	80.4	79.0	75.2	69.7	60.3	54.7	(29)
(30)			MCDB	55.0	59.2	69.8	74.6	82.7	87.6	90.2	88.8	85.4	76.5	65.7	57.5	(30)
(31)		5%	WB	45.8	49.4	57.9	63.3	70.8	76.2	78.9	77.2	73.1	66.7	57.4	49.1	(31)
(32)			MCDB	49.3	54.7	64.9	71.6	79.9	85.6	88.3	85.7	81.9	73.2	62.2	52.3	(32)
(33)		10%	WB	39.8	43.8	53.4	60.6	68.7	74.5	77.4	75.4	71.1	63.5	53.4	44.0	(33)
(34)			MCDB	43.3	48.9	59.4	69.0	76.7	83.2	85.8	82.9	79.2	70.8	59.0	47.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	16.4	18.8	21.0	19.9	19.2	19.4	19.7	23.1	21.2	17.6	14.5	(35)
(36)			MCDBR	21.6	23.7	24.3	26.2	22.6	21.4	20.7	21.4	24.9	25.1	22.6	19.8	(36)
(37)			MCWBR	17.6	17.9	15.0	14.3	10.3	9.4	9.5	9.5	10.5	12.7	15.5	16.1	(37)
(38)		5% WB	MCDBR	20.6	22.2	21.8	22.1	19.9	18.8	19.0	18.3	21.1	20.3	20.2	18.1	(38)
(39)	MCWBR		18.1	18.2	14.7	14.6	10.6	9.4	9.8	9.4	10.0	12.3	16.0	16.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.312	0.343	0.389	0.428	0.432	0.458	0.445	0.405	0.356	0.306	0.294	(40)	
(41)		taud	2.451	2.404	2.397	2.271	2.221	2.249	2.163	2.192	2.281	2.423	2.510	2.477	(41)	
(42)		Ebn at Noon	277	288	289	281	271	269	260	261	264	268	272	269	(42)	
(43)		Edh at Noon	25	30	34	41	44	43	47	44	38	30	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	597	868	1170	1535	1747	1997	1988	1754	1491	1028	688	502	(44)	
(45)		RadStd	69	80	86	104	140	154	122	112	105	116	64	64	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+178
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+135

Nomenclature: See separate page